

Integration by Substitution & Integration by Parts

MATH 146 – Calculus II

Torin Quinlivan

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The Fundamental Theorem of Calculus

Recall the Fundamental Theorem of Calculus from last lecture:

The Fundamental Theorem of Calculus

Suppose $f(t)$ is a continuous function on the interval $[a, b]$ and that $F(t)$ is an antiderivative of $f(t)$. Then

$$\int_a^b f(t) dt = F(b) - F(a)$$

The FTC says that you can evaluate $\int_a^b f(x) dx$ if you can find the antiderivative of $f(x)$. But how do we find $\int f(x) dx$?

Rules for Integrals

There is no “Product Rule” for integrals, so while it’s easy to find $\frac{d}{dx}f(x)g(x)$, it may not be as easy to find $\int f(x)g(x)$.

Remember that not all functions have clear antiderivatives, or antiderivatives which can be expressed in terms of elementary functions.

For example, the cumulative density function of the standard Normal distribution cannot be expressed easily and can generally only be found using numerical approximation methods.

$$\Phi(z) = \mathbb{P}[Z \leq z] = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^z e^{-\frac{t^2}{2}} dt$$

Rules for Integrals

We can, however, use the Chain Rule to give us a trick to help solve stubborn integrals.

Recall the Chain Rule:

$$\frac{d}{dx} f(g(x)) = f'(g(x)) g'(x)$$

Then we can have:

$$\int f'(g(x)) g'(x) dx = \int \frac{d}{dx} f(g(x)) dx = f(g(x))$$

Integration by Substitution

Integration by Substitution

$$\int f'(g(x)) g'(x) dx = \int f(u) du,$$

where

$$u = g(x).$$

Example: Integration by Substitution

Example: $\int 2xe^{x^2} dx.$

Example: Integration by Substitution

Example: Find $\int (x + 1)\sqrt{3x + 2} \, dx$.

Example: Find $\int \frac{1}{4x + 3} \, dx$.

Integration by Substitution for Definite Integrals

Integration by Substitution for Definite Integrals

$$\int_a^b f'(g(x)) g'(x) dx = \int_{g(a)}^{g(b)} f(u) du,$$

Example: Find $\int_0^{\pi/2} \sin^2(x) \cos(x) dx$.

Example: Find $\int_0^1 e^{-x} (2e^{-x} + 3)^9 dx$.

Integration by Parts

There is something similar to the Product Rule we can use, by reversing engineering the Product Rule.

Recall the Chain Rule:

$$\frac{d}{dx} f(x)g(x) = f(x)g'(x) + f'(x)g(x)$$

Integration by Parts

Then we can have:

$$f(x)g(x) = \int (f(x)g(x))' dx$$

$$f(x)g(x) = \int f(x)g'(x) + f'(x)g(x) dx$$

$$f(x)g(x) = \int f(x)g'(x) dx + \int f'(x)g(x) dx$$

$$\int f(x)g'(x) dx = f(x)g(x) - \int f'(x)g(x) dx$$

$$\int f(x) \left(\frac{dg}{dx} \right) \cancel{dx} = f(x)g(x) - \int g(x) \left(\frac{df}{dx} \right) \cancel{dx}$$

$$\int f(x) dg = f(x)g(x) - \int g(x) df$$

Integration by Parts

Integration by Parts

$$\int u \, dv = uv - \int v \, du$$

Example: Integration by Parts

Example: $\int x \sin(x) \, dx.$

Example: $\int x \ln(x) \, dx.$

Example: Integration by Parts

Example: $\int \ln(x) \, dx.$

Example: Show that $\int (\ln(x))^n \, dx = x (\ln(x))^n - n \int (\ln(x))^{n-1} \, dx.$